Lecture 4: Evaluating Models

- DS-204 Introduction to Machine Learning, winter term 2026/27
- Example University (fictional course, sample notes for testing)
- Lecturer: Dr. A. Example
- Today: train/test split, cross-validation, confusion matrix, precision, recall, F1, ROC/AUC, regression metrics

Train/test split

- We hold back data the model never sees during training: the test set.
- A typical split is 80% training data and 20% test data.
- Never tune hyperparameters on the test set. Use a separate validation set instead.
- The test set is used only once, at the very end, to report the final score.

Cross-validation

- k-fold cross-validation splits the data into k equal parts (folds).
- Each fold is used once as validation data while the other k-1 folds train the model.
- Common choices are k = 5 or k = 10.
- Stratified k-fold keeps the class proportions the same in every fold. Use it for imbalanced classes.

The confusion matrix

- True positive (TP): predicted positive, actually positive.
- False positive (FP): predicted positive, actually negative. Also called a type I error.
- False negative (FN): predicted negative, actually positive. Also called a type II error.
- True negative (TN): predicted negative, actually negative.

Precision and recall

- Precision = TP / (TP + FP): of everything we flagged, how much was right?
- Recall = TP / (TP + FN): of everything we should have found, how much did we find?
- Example spam filter: precision 0.9 means 9 out of 10 emails marked as spam really are spam.
- The same filter has a recall of 0.6: it catches 6 out of 10 spam emails.

The F1 score

- F1 is the harmonic mean of precision and recall: F1 = 2 * P * R / (P + R).
- It is only high when both precision and recall are high.
- Spam filter example: F1 = 2 * 0.9 * 0.6 / (0.9 + 0.6) = 0.72.
- Use F1 instead of accuracy when the classes are imbalanced.

Overfitting and underfitting

- Overfitting: very high training accuracy but much lower test accuracy. The model memorised noise.

- Underfitting: low accuracy on both training and test data. The model is too simple.

- Remedies for overfitting: more training data, L1 or L2 regularization, early stopping, dropout.

- Remedies for underfitting: a more complex model or better features.

ROC curve and AUC

- The ROC curve plots the true positive rate against the false positive rate for every threshold.
- AUC is the area under the ROC curve.
- AUC = 0.5 means the model is no better than random guessing.
- AUC = 1.0 means the model separates the two classes perfectly.

Metrics for regression

- MAE (mean absolute error): the average size of the errors.
- MSE (mean squared error): the average of the squared errors.
- RMSE is the square root of MSE and has the same unit as the target.
- RMSE punishes large errors more than MAE, because errors are squared before averaging.

Exam information

- The written exam for DS-204 takes place on 12 February 2027.
- It lasts 90 minutes.
- You may bring one handwritten A4 sheet with notes (both sides). No calculators.
- Topics: lectures 1 to 6, with a focus on model evaluation.